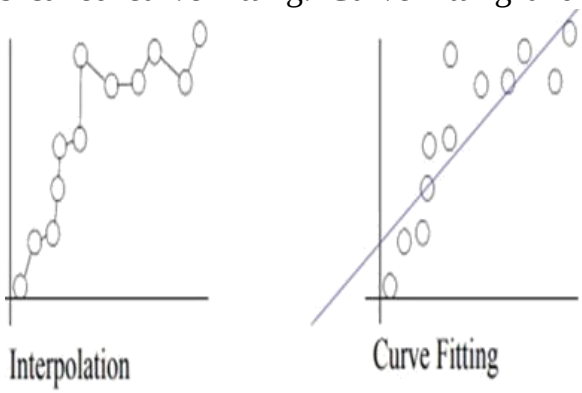
Chapter 4

Curve Fitting & Correlation

4.1 Introduction

The process of constructing an approximate curve $y = f(x)$, which fit best to a given discrete set of points $(x_i, y_i),\ i = 1, 2, 3, \dots, n$ is called curve fitting. Curve fitting and interpolation are closely associated procedures. In interpolation, the fitted function should pass through all given data points; whereas curve fitting methodologically fits a unique curve to the data points, which may or may not lie on the fitted curve. The difference between interpolation and curve fitting; while attempting to fit a linear function; is illustrated in the adjoining figure.

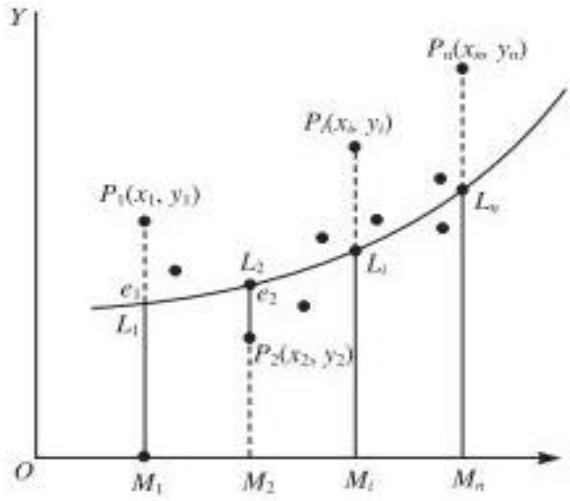


4.2 Principle of Least Squares

The principle of least squares is one of the most popular methods for finding the curve of best fit to a given data set $(x_i, y_i),\ i = 1, 2, 3, \dots, n$.

Let $y = f(x)$ be the equation of the curve to be fitted to the given set of points $P_1(x_1, y_1)$, $P_2(x_2, y_2), \cdots, P_n(x_n, y_n)$.

Then at a point $x = x_i$, the observed value of the ordinate is $P_iM_i = y_i$ say and let the expected (theoretical) value be $f(x_i)$, shown by L_iM_i in the adjoining figure. The difference between the observed and expected values is the error $e_i = P_iL_i$



Then $e_1 = y_1 - f(x_1)$

$e_2 = y_2 - f(x_2)$

$\vdots$

$e_n = y_n - f(x_n)$

Squaring each error e_i (to take care of negative errors) and adding, we get

$E = e_1^2 + e_2^2 + \cdots . + e_n^2 = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - f(x_i))^2$. The curve of best fit is that for which E is minimum. This is called the Principle of least squares.

4.2.1 Fitting a straight line $y = ax + b$.

Let $y = ax + b$ be the straight line to be fitted to the given set of data points (x_1, y_1), (x_2, y_2),……, (x_n, y_n).

Then $e_i = y_i - f(x_i) = y_i - (ax_i + b), \quad i = 1, 2, \ldots, n$

$$\therefore E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \big(y_i - (ax_i + b)\big)^2$$

Now by the principle of least square, for the curve of best fit, E is minimum

$$\therefore \frac{\partial E}{\partial a} = 0 \text{ and } \frac{\partial E}{\partial b} = 0$$

$$\therefore \frac{\partial E}{\partial a} = 0 \qquad \text{and} \qquad \frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \sum_{i=1}^{n} 2\left(y_i - (ax_i + b)\right)(-x_i) = 0$$

$$\Rightarrow \sum_{i=1}^{n} (y_i x_i - ax_i^2 - bx_i) = 0$$

$$\Rightarrow \sum_{i=1}^{n} y_i x_i = a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i \quad \ldots ①$$

$$\Rightarrow \sum_{i=1}^{n} 2(y_i - ax_i - b)(-1) = 0$$

$$\Rightarrow \sum_{i=1}^{n} (y_i - ax_i - b) = 0$$

$$\Rightarrow \sum_{i=1}^{n} y_i = a\sum_{i=1}^{n} x_i + nb \quad \ldots ②$$

Since x_i, y_i are known set of values, equation ① and ② are two equation with variables a and b, ignoring the suffices, ② and ① can be rewritten as

$$\sum y = a\sum x + nb \quad \ldots ③$$

and $$\sum xy = a\sum x^2 + b\sum x \quad \ldots ④$$

③and ④are known as Normal equation for fitting a straight line $y = ax + b$

- If the equation of line is taken as $y = a + bx$
  we get normal equations as: $\sum y = na + b\sum x$ and $\sum xy = a\sum x + b\sum x^2$

Example 1 By the method of least squares, find a straight line that best fits the following data points.

x	0	1	2	3	4
y	1.0	2.9	4.8	6.7	8.6

Solution: Let line of best fit be given by $y = ax + b$ ... ①

Normal equations are given by:

$$\sum y = a\sum x + nb \quad \ldots ②$$

and $$\sum xy = a\sum x^2 + b\sum x \quad \ldots ③$$

Calculating $\sum x, \sum y, \sum xy$ and $\sum x^2$

x	y	xy	x^2
0	1.0	0	0
1	2.9	2.9	1
2	4.8	9.6	4
3	6.7	20.1	9
4	8.6	34.4	16
$\sum x = 10$	$\sum y = 24$	$\sum xy = 67.0$	$\sum x^2 = 30$

Substituting values of $\sum x, \sum y, \sum xy$ and $\sum x^2$ in ② and ③

$$\Rightarrow 24 = 10a + 5b \quad \text{...④}$$

and $$67 = 30a + 10b \quad \text{...⑤}$$

Solving ④and ⑤, we get $a = 1.9$ and $b = 1$

Substituting in ①, line of best fit is $y = 1.9x + 1$

Example 2 Fit a straight line to following data

x	0	1	2	3	4
y	1.0	1.8	3.3	4.5	6.3

Solution: Let line of best fit be given by $y = ax + b$ ...①

Normal equations are given by:

$$\sum y = a\sum x + nb \quad \text{...②}$$

and $$\sum xy = a\sum x^2 + b\sum x \quad \text{...③}$$

Calculating $\sum x, \sum y, \sum xy$ and $\sum x^2$

x	y	xy	x^2
0	1.0	0	0
1	1.8	1.8	1
2	3.3	6.6	4
3	4.5	13.5	9
4	6.3	25.2	16
$\sum x = 10$	$\sum y = 16.9$	$\sum xy = 47.1$	$\sum x^2 = 30$

Substituting values of $\sum x, \sum y, \sum xy$ and $\sum x^2$ in ② and ③

$$\Rightarrow 16.9 = 10a + 5b \quad \text{...④}$$

and $47.1 = 30a + 10b$ ... ⑤

Solving ④and ⑤, we get $a = \frac{133}{100} = 1.33$ and $b = \frac{18}{25} = 0.72$

Substituting in ①, line of best fit is $y = 1.33x + 0.72$

Example 3 If F is the force required to lift a load W, by means of a pulley, fit a linear expression $F = a + bW$ against the following data:

W	50	70	100	120
F	12	15	21	25

Solution: Line for best fit is given as $F = a + bW$ ... ①

Normal equations are given by:

$\sum F = na + b\sum W$ ... ②

and $\sum WF = a\sum W + b\sum W^2$ ... ③

W	F	WF	W^2
50	12	600	2500
70	15	1050	4900
100	21	2100	10000
120	25	3000	14400
$\sum W = 340$	$\sum F = 73$	$\sum WF = 6750$	$\sum W^2 = 31800$

Substituting values of $\sum W, \sum F, \sum WF$ and $\sum W^2$ in ② and ③

$\Rightarrow 73 = 4a + 340b$ ... ④

and $6750 = 340a + 31800b$ ... ⑤

Solving ④and ⑤, we get $a = 2.2759$ and $b = 0.1879$

Substituting in ①, line of best fit is $F = 2.2759 + 0.1879W$

4.2.2 Fitting a parabola $y = ax^2 + bx + c$

Let $y = ax^2 + bx + c$ be the parabola to be fitted to the given set of data points $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$.

Then $e_i = y_i - f(x_i) = y_i - (ax_i^2 + bx_i + c)$, $i = 1, 2, \ldots, n$

$\therefore E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - (ax_i^2 + bx_i + c)\right)^2$

Now by the principle of least square, for the curve of best fit, E is minimum

$\therefore \frac{\partial E}{\partial a} = 0$, $\frac{\partial E}{\partial b} = 0$ and $\frac{\partial E}{\partial c} = 0$

Solving we get normal equations as:

$\sum y = a\sum x^2 + b\sum x + nc$

$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$

$\sum x^2 y = a\sum x^4 + b\sum x^3 + c\sum x^2$

Example 4 Fit a parabola $y = ax^2 + bx + c$ to the given data

x	10	12	15	23	20
y	14	17	23	25	21

Solution: Let the parabola of best fit be given by $y = ax^2 + bx + c$ ... ①

Normal equations are given by:

$\sum y = a\sum x^2 + b\sum x + nc$ ... ②

$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$ ... ③

$\sum x^2 y = a\sum x^4 + b\sum x^3 + c\sum x^2$ ... ④

x	y	xy	x^2	x^2y	x^3	x^4
10	14	140	100	1400	1000	10000
12	17	204	144	2448	1728	20736
15	23	345	225	5175	3375	50625
23	25	575	529	13225	12167	279841
20	21	420	400	8400	8000	160000
$\sum x =$ **80**	$\sum y =$ 100	$\sum xy =$ 1684	$\sum x^2 =$ 1398	$\sum x^2y =$ 30684	$\sum x^3 =$ 26270	$\sum x^4 =$ 521202

Substituting values of $\sum x, \sum y, \sum xy$ and $\sum x^2$ in ② and ③ and ④

$\Rightarrow 100 = 1398a + 80b + 5c$ ... ⑤

$1684 = 26270a + 1398b + 80c$ ... ⑥

$30648 = 521202a + 26270b + 1398c$ ... ⑦

Solving ⑤ ⑥ and ⑦ , we get $a = -0.07$, $b = 3.01$, $c = -8.73$

Substituting in ①, parabola of best fit is $y = -0.07\,x^2 + 3.01x - 8.73$

Example 5 Fit a 2nd parabola to the given data

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

Solution: Let the parabola of best fit be given by $y = ax^2 + bx + c$ ... ①

x	y	xy	x^2	x^2y	x^3	x^4
1	1	1	1	1	1	1
3	2	6	9	18	27	81
4	4	16	16	64	64	256
6	4	24	36	144	216	1296
8	5	40	64	320	512	4096
9	7	63	81	567	729	6561
11	8	88	121	968	1331	14641
14	9	126	196	1764	2744	38416
$\sum x =$ 56	$\sum y =$ 40	$\sum xy =$ 364	$\sum x^2 =$ 524	$\sum x^2y =$ 3846	$\sum x^3 =$ 5624	$\sum x^4 =$ 65348

Normal equations are given by:

$$\sum y = a\sum x^2 + b\sum x + nc \quad \text{...②}$$

$$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x \quad \text{...③}$$

$$\sum x^2y = a\sum x^4 + b\sum x^3 + c\sum x^2 \quad \text{...④}$$

Substituting values of $\sum x, \sum y, \sum xy$ and $\sum x^2$ in ② and ③ and ④

$$\Rightarrow 40 = 524a + 56b + 8c \quad \text{...⑤}$$

$$364 = 5624a + 524b + 56c \quad \text{... ⑥}$$

$$3846 = 65348a + 5624b + 524c \quad \text{... ⑦}$$

Solving ⑤ ⑥ and ⑦ , we get

$a = \frac{103}{11229} = 0.009$, $b = \frac{8672}{11229} = 0.77$, $c = \frac{4375}{22458} = 0.195$

Substituting in ①, parabola of best fit is $y = 0.009\,x^2 + 0.77x + 0.195$

4.2.3 Change of Scale

If the data values are equispaced (with height (h)) and quite large for computation, simplification may be done by origin shifting as given below:

- When number of observations (n) is odd, take the origin at middle value of the table; say (x_0) and substitute $u = \frac{x - x_0}{h}$
- y values if small; may be left unchanged; or we can shift them at average value of y data $v = \frac{y - y_0}{h}$